Data Acquisition & Market Sources

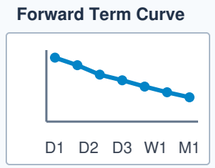
Physical Grid Fundamentals (REE)

- 16 continuous grid features (hourly to daily)
- Generation mix: wind, solar, hydro, CC
- Grid demand & generation imbalance



OMIP Derivatives & OMIE Spot

- 14 tradable contracts (Baseload SPEL)
- Futures: D1–D3, W1–W4, M1–M6
- OMIClear daily margin clearing



Macroeconomic News Feeds

- 1,242 consecutive market days (2023–2026)
- Comprehensive press: Reuters, Montel, etc.
- Gas supply, EU ETS carbon & geopolitics

LLM News Sentiment & Feature Engineering

1. Zero-Shot LLM Sentiment (Llama-3 8B)

- Local Ollama inference (zero internet / no data leakage)
- Numeric structured sentiment in [-1.0, +1.0]

φ_1–φ_3: Short/Med/Long Outlook	φ_4: LNG & Natural Gas
φ_5: EU ETS CO_2 Policy	φ_6: MIBEL Regulation
φ_7: Generation Mix / Hydro	φ_8–φ_{10}: Geopolitics & Macro

2. Causal Feature Engineering Pipeline

1. **Causal Lags Enforcement**
   Features conditioned only on data published up to t–1.
2. **Temporal Imputation**
   Forward-filling weekend quotes with NO_TRADE masks.
3. **Rolling Standardization**
   Z-score scalers fit strictly on historical training slice D_train.
4. **Orthogonal PCA Structure**
   Confirms sentiment signals add orthogonal variance to grid.

Feature Representations & Factorial Ablation

Strict 2×2 Factorial Ablation

Base (X_Common)
- Dataset_Exogenous_Calendar
- Physical + OMIP + Calendar
- Dim: 35 features

Quant (X_Quant)
- Dataset_Exogenous_Commodities
- Common (35) + 7 commodities
- Dim: 42 features

Hybrid (X_Hybrid)
- Dataset_M8_Physical_NLP
- Common (35) + 10 sentiment
- Dim: 45 features

Full (X_Full)
- Dataset_NLP_Subgroups
- Common + Commodities + NLP
- Dim: 52 features

Controlled Comparisons

- **NLP vs Quant: Full vs Quant +1.59 pp (2025)**
- **NLP vs Base: Hybrid vs Common +4.18 pp (2025)**
- **Text vs Commodities: Hybrid better during 2026 shock**
- **Noise robustness: Curated vs uncurated news**

DRL Model Formulation & Walk-Forward Retraining

1. Markov Decision Process (MDP) Procurement

- Action space A: 14 discrete contracts (SPOT, D1–D3, W1–W4, M1–M6)
- Observation o_t: 7-day lagged state + 400-day forward inventory queue c_t
- Reward r_t = (P_SPOT,t+1 – P_eff,t+1) – R_penalty · I_invalid
- Clearing rules: OMIClear variation margins + OMIE physical delivery

2. Deep Recurrent & Attention Architectures

RecurrentPPO (LSTM)
- Hidden state h_t (d_h = 256 / 64)
- Generalized Advantage Estimation (GAE)

GTrXL Transformer
- Multi-head attention (H=4, L=2, M=14)
- GRU skip connections & LayerNorm

3. Continuous Walk-Forward Protocol (B = 41 blocks)

- Rolling update horizons W ∈ {1, 3, 6, 12} months across 2023–2026
- Bounded compute budget (K = 15,000 steps/month; warm-start updates)
- Multi-seed robustness: 5 independent evaluation seeds (10, 20, 30, 40, 50)

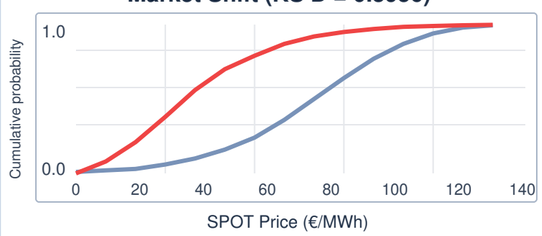
Train: $M_{(b-1)}$ (W) → Out-of-Sample: M_β → Persist inventory c_eval → $M_{(b+1)}$

Performance & Downside Risk Mitigation (2026 Dislocation)

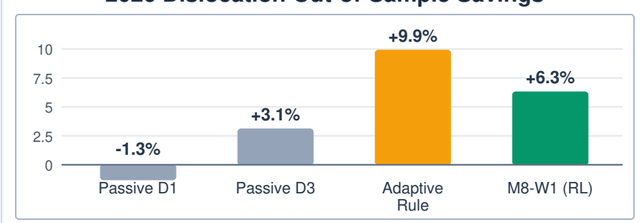
1. Core Empirical Findings & Downside Risk Protection

- +6.32% multi-seed savings under 2026 wholesale deflation (SPOT €44.22/MWh).
- Mitigated maximum drawdown: reduced by ~66% vs D3 (186.45 vs 548.95 €/MWh).
- Lower downside risk: σ_- = 9.15 €/MWh (vs 13.68 for D2 and 18.63 for D3).
- Dynamic multi-horizon allocation: hedging across D2, D1, SPOT, and D3.

Market Shift (KS D = 0.3956)



2026 Dislocation Out-of-Sample Savings



2. SHAP Explainability & Macroeconomic Grounding

- Attribution shows the policy uses news sentiment.
- Leading signals: regulatory risk (φ_6), LNG (φ_4), generation mix (φ_7).
- Policy avoids overpriced long-dated contracts in falling markets.
- Savings persist with fees up to 0.25 €/MWh across 5 seeds.

